

Timesheet reconciliation

Use cases (2) — A1–A11 — run against reconcile.py

Settings: sensitivity 10m, auto-correct ceiling 60m, ignored GPS gap 5m, ignored stop 9m (dither stops in the B commute are absorbed; B16’s 10-minute onsite visit is not).

ID	Deck	Code	Verdict	
A1	AUTO-EDIT	AUTO-EDIT	MATCH	The first drive of the day ran long compared to booked working time
A2	AUTO-EDIT	AUTO-EDIT	MATCH	The last drive to home started before the technician stopped logging work
A3	DISCREPANCY	DISCREPANCY	MATCH	Travel time between two jobs was recorded as job work.
A4	DISCREPANCY	DISCREPANCY	MATCH	Materials at the office, the drive, and the on-site work as one block
A5	DISCREPANCY	DISCREPANCY	MATCH	The technician left the job and came back
A6	DISCREPANCY	DISCREPANCY	MATCH	Driving was logged while the vehicle never moved
A7	NO ACTION	NO ACTION	MATCH	A four-minute mismatch is noise, not a finding
A8	DISCREPANCY	DISCREPANCY	MATCH	The same correction as A1 — but two and a quarter hours of it
A9	REVIEW FLAG	DISCREPANCY	SAME TIMES, DIFFERENT LABEL	The van was on site longer than the timesheet claims
A11	NO ACTION	NO ACTION	MATCH	The fleet feed has a hole in the middle of the job

Notes

A9 is the only leftover: the deck’s review flag is reported as DurationMismatch. Times match.

Auto-edit is a cutover moved onto a driving timesheet. Parked time offered to working, time with nobody to take it, and unlogged time around a timesheet are flagged; the suggested span is in the discrepancy intervals.

Timesheet reconciliation

Use cases (3) — B1–B16 — same engine, same settings

CODE OUTPUT splits each timesheet into the original kept span (solid) versus minutes the engine drops (peach) or adds (dashed). INTERVALS is Result.intervals: the vehicle on the suggested span. A discrepancy leaves logged times in place.

ID	Deck	Code	Verdict	
B1	DISCREPANCY	DISCREPANCY	MATCH	Working starts while the van is still offsite; first onsite stop is later
B2	DISCREPANCY	DISCREPANCY	MATCH	Logged gap before working; first onsite stop is still later
B3	DISCREPANCY	DISCREPANCY	MATCH	Drive ended early and working ended before the van left the site
B4	DISCREPANCY	DISCREPANCY	MATCH	Driving was logged after the van had already arrived on site
B5	DISCREPANCY	DISCREPANCY	MATCH	Driving ran into the site, and working also ended early
B6	AUTO-EDIT	AUTO-EDIT	DIFFERS	Drive ended early; working already covers the onsite window
B7	DISCREPANCY	NO ACTION	SAME TIMES, DIFFERENT LABEL	The van never reached Job A
B8	AUTO-EDIT	AUTO-EDIT	DIFFERS	First drive starts with the fleet; cutover is early; working covers the visit
B9	DISCREPANCY	DISCREPANCY	MATCH	Driving ran 12 minutes into the onsite stop
B10	AUTO-EDIT	AUTO-EDIT	DIFFERS	An 11-minute early cutover, working covers the visit
B11	DISCREPANCY	DISCREPANCY	MATCH	Driving ran 14 minutes into the onsite stop
B12	DISCREPANCY	DISCREPANCY	MATCH	Meal break between two working timesheets at the same job
B13	DISCREPANCY	DISCREPANCY	MATCH	A drive between two working timesheets at the same job
B14	DISCREPANCY	NO ACTION	SAME TIMES, DIFFERENT LABEL	Working was logged while the van sat offsite
B15	DISCREPANCY	NO ACTION	SAME TIMES, DIFFERENT LABEL	Commute correctly not claimed
B16	DISCREPANCY	DISCREPANCY	MATCH	A ten-minute visit, then driving away still logged as work

Notes

B4/B9/B11/B12 now clear the 10m floor and match as discrepancies. B10 auto-edits the 11m cutover. B6/B8/B10 still differ on the 8-minute trailing drive-away (under the floor, and not a hand-over onto driving).

B7 and B14 never match a job site, so the day is skipped. B15’s unclaimed commutes are not invented as timesheets.

A1

The first drive of the day ran long compared to booked working time

The technician stamped a short drive and then went straight into working at Job A. The vehicle was still moving for another half hour.



Day total unchanged: 3h00 logged, 3h00 after. Minutes moved onto the driving timesheet that already existed.

CODE MATCH

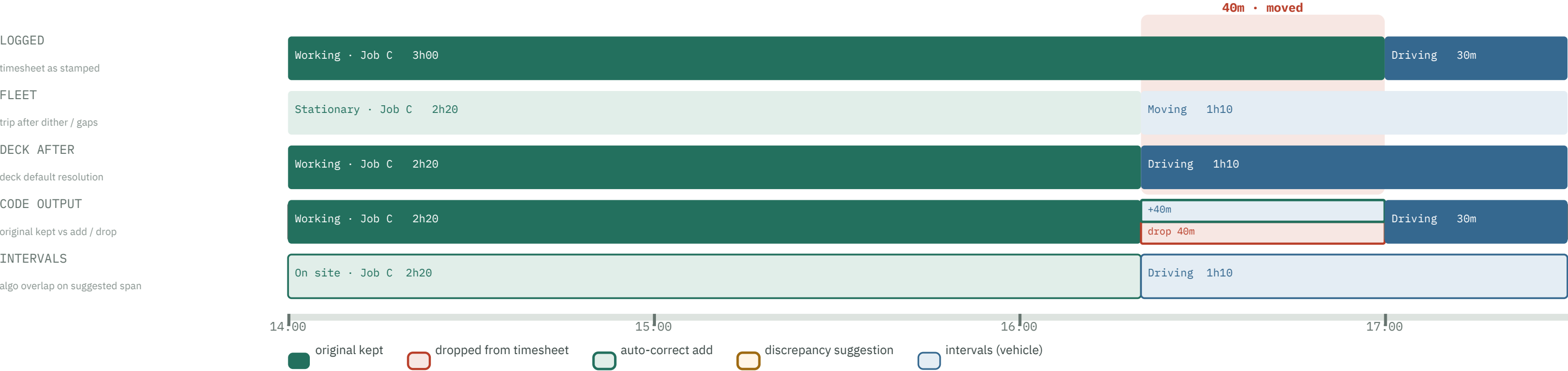
Working starts while the van is still moving. That leading stretch contradicts a working timesheet, so the 30 minutes are handed to the neighbouring driving timesheet (which also reaches right into the same trip). Change is 30m, under the 60m ceiling, so both sides auto-edit. Day total stays 3h00.

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Extend driving	30m	AUTO-EDIT	AUTO-EDIT

A2

The last drive to home started before the technician stopped logging work

Job C was stamped closed at 17:00, but the van pulled away from the site at 16:20 and the drive-home timesheet only starts at 17:00.



AUTO-EDIT

deck

AUTO-EDIT

code

40m · moved

Both sides of the shared cutover move together — 3h30 logged, 3h30 after.

CODE MATCH

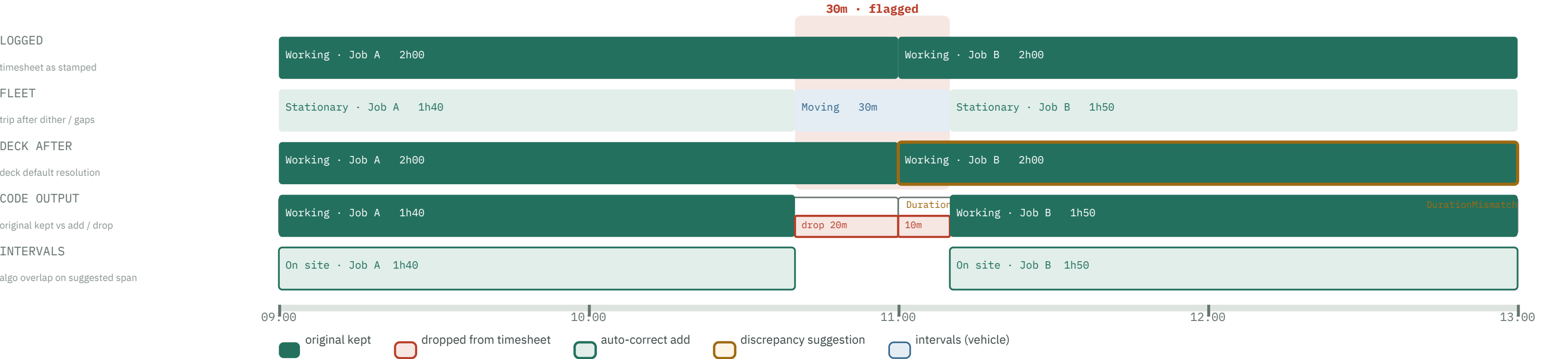
Mirror of A1 at the end of the day. Working's trailing edge is moving; driving is the only neighbour that can own it. 40m is handed across the cutover. Both sides auto-edit; 3h30 logged, 3h30 after.

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Extend driving	40m	AUTO-EDIT	AUTO-EDIT

A3

Travel time between two jobs was recorded as job work.

Job A is followed immediately by Job B, with no driving time recorded between them. Fleet data shows a 30-minute drive between the two jobs.



Nothing changes until a human decides. There is no driving timesheet to receive the minutes, and we do not invent one.

CODE MATCH

Job A's trailing 20 minutes are moving, which contradicts a working timesheet, but the only neighbour is another working timesheet, so there is nobody to hand them to. A slice with no driving taker is never applied: Job A is flagged DurationMismatch and left as logged, with the suggestion (end at 10:40) carried by its intervals. Job B's 11:00–11:10 head is 10m, at the floor, so it is flagged too.

DISCREPANCY

deck

DISCREPANCY

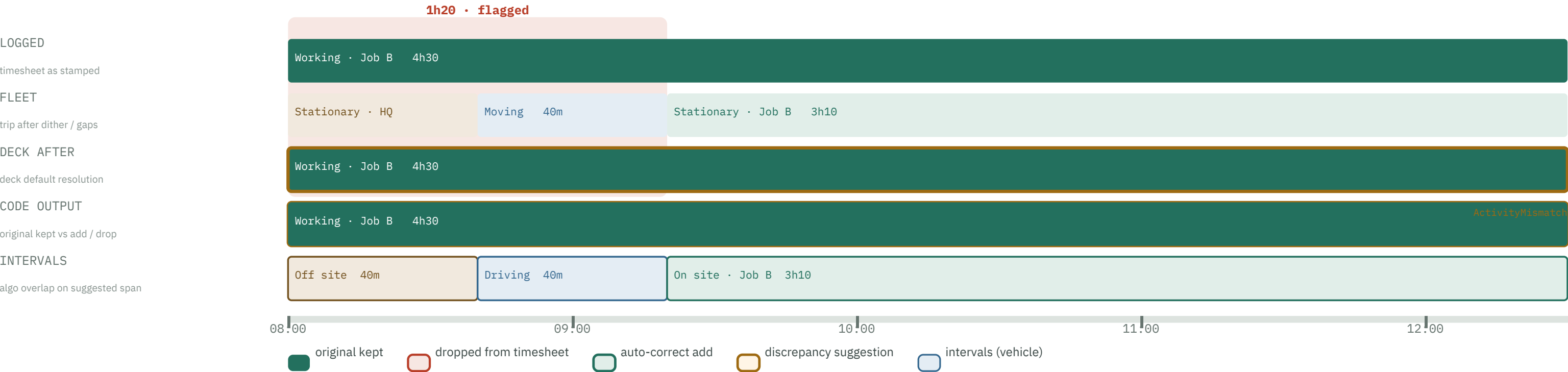
code

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Company-dependent	30m	DISCREPANCY	DISCREPANCY

A4

Materials at the office, the drive, and the on-site work as one block

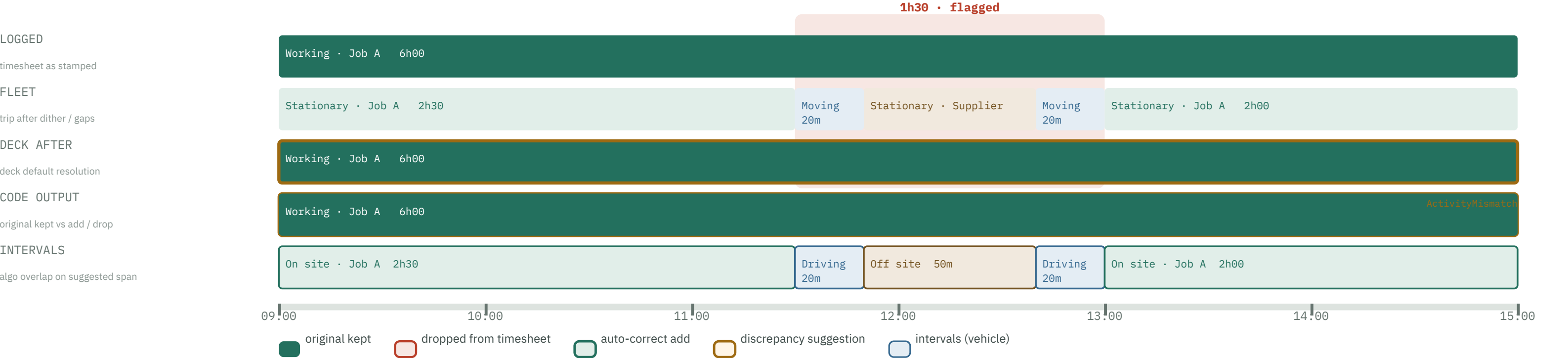
One Job B timesheet includes the materials pickup at HQ, the drive to the job, and the on-site work. Fleet data shows them as three separate activities.



A5

The technician left the job and came back

One six-hour working timesheet at Job A. The van left the site for ninety minutes in the middle, stopping at a supplier.



The visit is never auto-split. Splitting it would create two timesheets and force a guess about the gap.

CODE MATCH

First and last acceptable trips are the two Job A stops. The 90-minute away stretch in the middle — drive, supplier, drive — is significant and not acceptable, so DurationMismatch, timesheet not split. Same outcome as the deck.

DISCREPANCY

deck

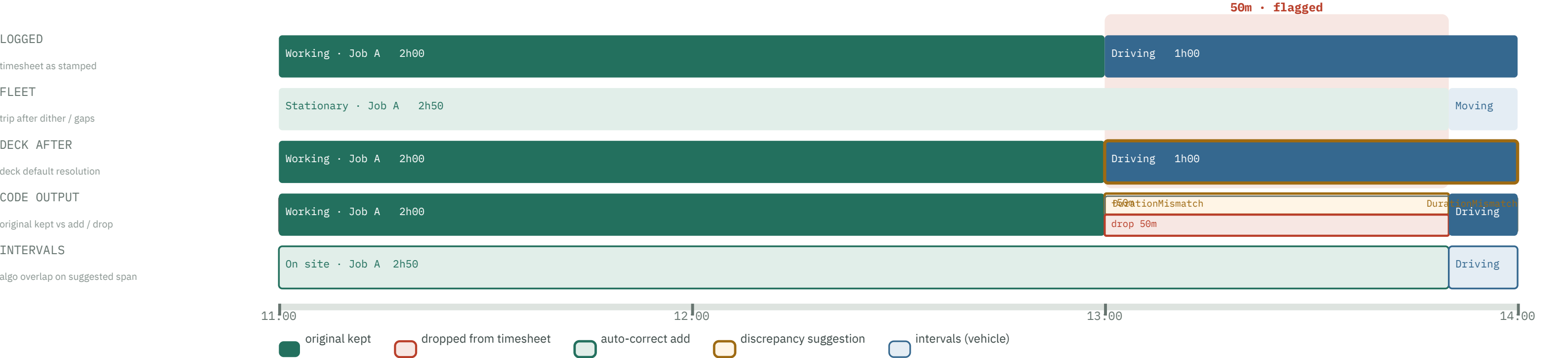
DISCREPANCY

code

A6

Driving was logged while the vehicle never moved

The driving timesheet starts at 13:00, but fleet data shows the van parked at Job A until 13:50. 13:00–13:50 was incorrectly recorded as driving.



Suggested move: give the 50 minutes to the working timesheet at Job A — offered, not applied. Parked at Job A is not treated as proof of working.

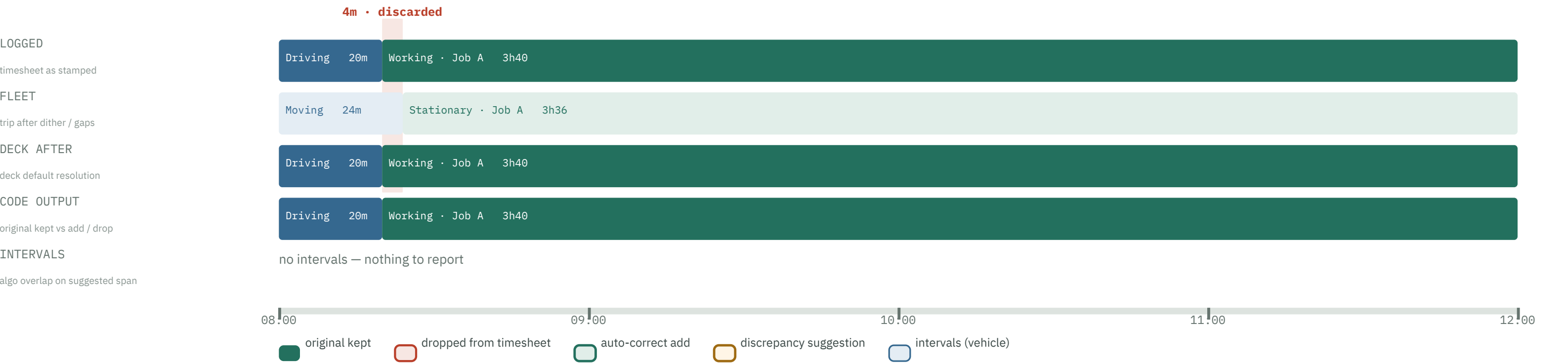
CODE MATCH

Driving's 50-minute head is parked at Job A, which contradicts a driving timesheet, and the working neighbour could take it. But parked is not proof of working: only a hand-over onto a driving timesheet is applied on the engine's authority, so the move is offered, not applied. Both timesheets are flagged DurationMismatch; their intervals show the suggestion (Working 11:00–13:50, Driving 13:50–14:00) while the times stay as logged.

A7

A four-minute mismatch is noise, not a finding

The driving timesheet ends four minutes before the van stops. The difference is real, but too small to correct.



Below the sensitivity floor (10m): discarded outright. No edit, and no discrepancy either.

CODEMATCH

Working would hand 08:20–08:24 to driving, but that slice is 4m and is discarded by the 10m sensitivity floor. No remaining operations, no status. Empty output: no action.

NO ACTION

deck

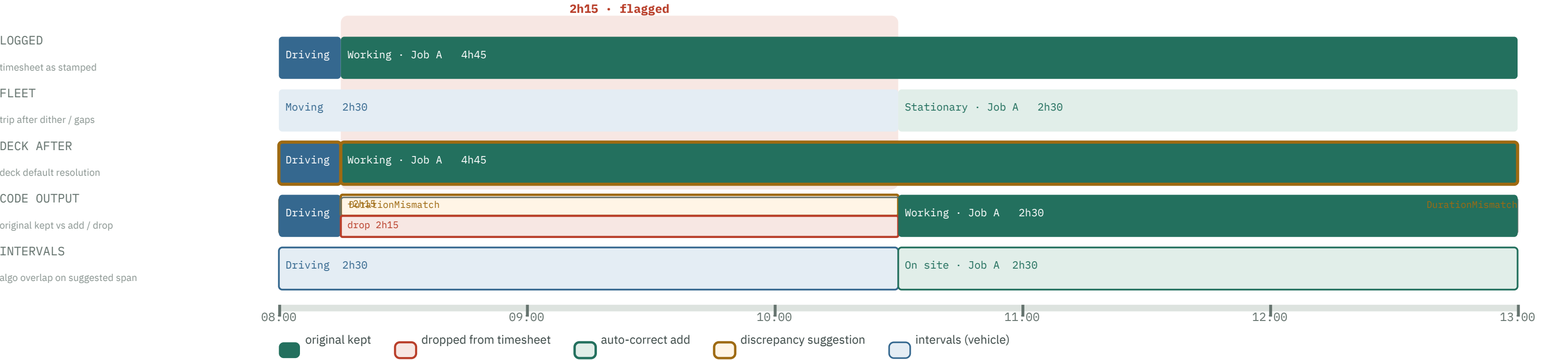
NO ACTION

code

A8

The same correction as A1 — but two and a quarter hours of it

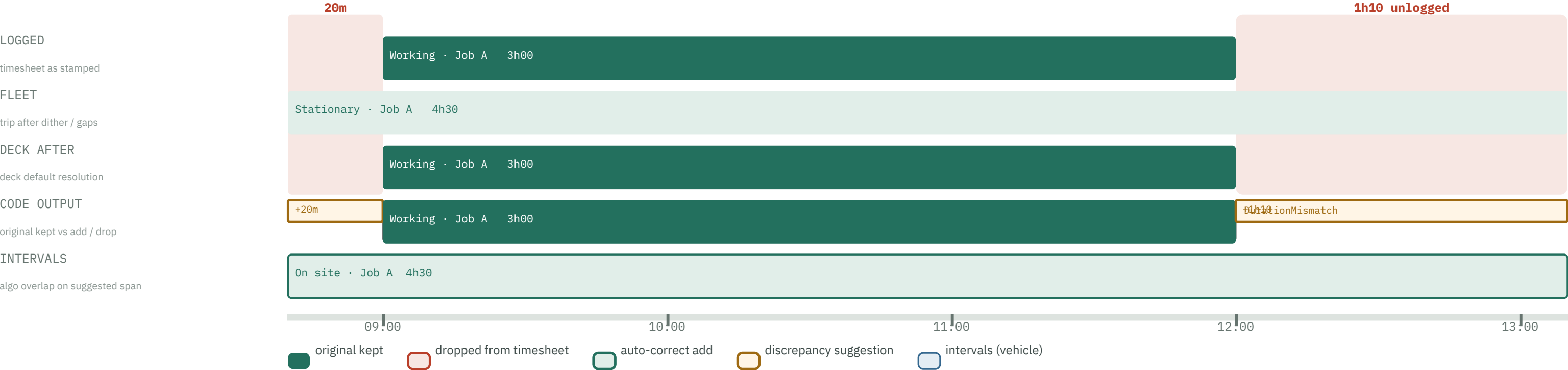
Identical shape to the first scenario: an understated first drive. The size is what changes the answer.



A9

The van was on site longer than the timesheet claims

It is possible — the working timesheet sits entirely inside the on-site window. There is simply more parked time around it.



Flagged for review only. Parked at the job is not proof of work. Extra on-site time is never auto-applied.

CODE

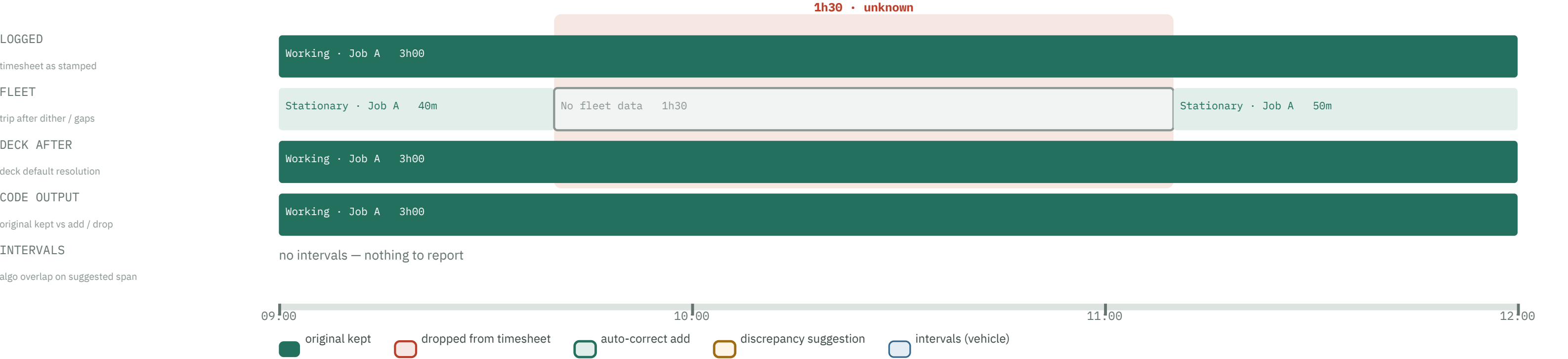
SAME TIMES, DIFFERENT LABEL

The van sat at Job A from 08:40 to 13:10; the timesheet claims 09:00–12:00. Extra on-site time is evidence for a reviewer, never an auto-edit: the two reach slices (20m before, 1h10 after) flag the timesheet as DurationMismatch and the times stay as logged. The intervals report the whole on-site stay, 08:40–13:10, so the reviewer sees what the flag is about. The code has no separate review-flag label; DurationMismatch is that flag.

A11

The fleet feed has a hole in the middle of the job

Ninety minutes with no usable position or motion data. The log claims work at Job A the whole time.



Absence of evidence is not a contradiction. Gaps are not detected, not edited, and not raised.

CODE MATCH

A 90-minute GPS hole is not a contradiction. Internally this is IncompleteGps: the timesheet is neither corrected nor reported, and the output is empty. No action, matching the deck.

NO ACTION

deck

NO ACTION

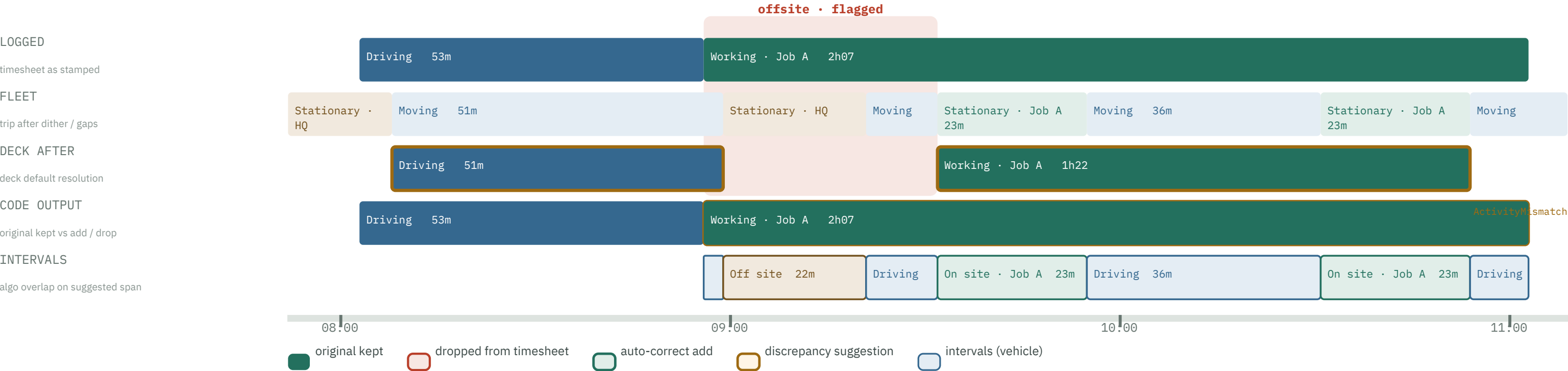
code

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Not a contradiction	Not asked	—	NO ACTION	NO ACTION

B1

Working starts while the van is still offsite; first onsite stop is later

Logged driving then working at Job A, back-to-back. Fleet shows a chopped commute, an offsite stop, then two onsite visits with driving in between.



Default resolution moves the cutover to the first onsite stop and trims working to the last onsite stop, leaving a gap / idle in between.

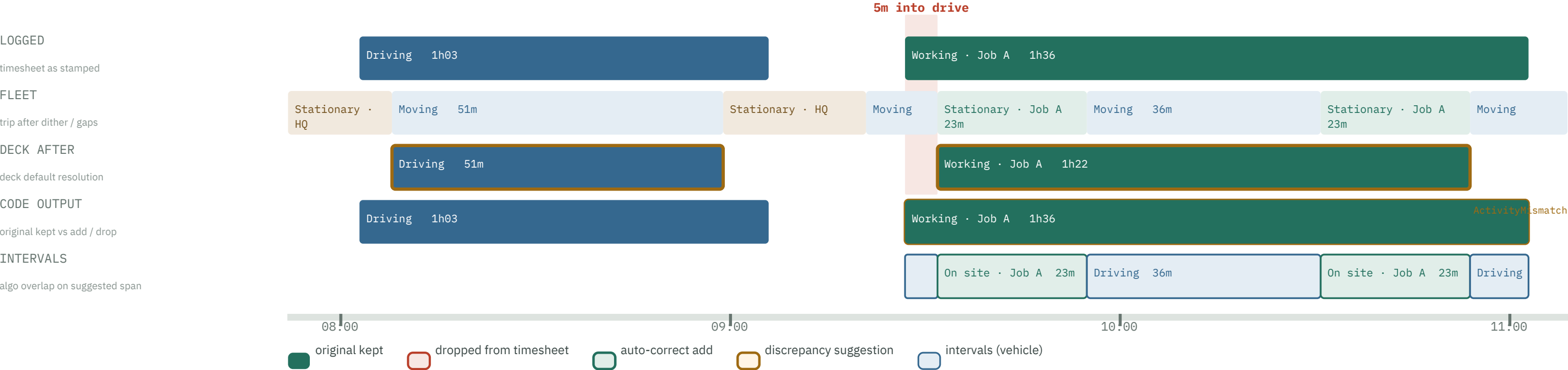
CODE MATCH

Dither on the commute is absorbed into one moving trip 08:08–08:59. Working then overlaps offsite time, a drive, two onsite stops and more driving — all significant — so DurationMismatch. Times stay as logged. The suggestion in the intervals starts at the first onsite stop.

B2

Logged gap before working; first onsite stop is still later

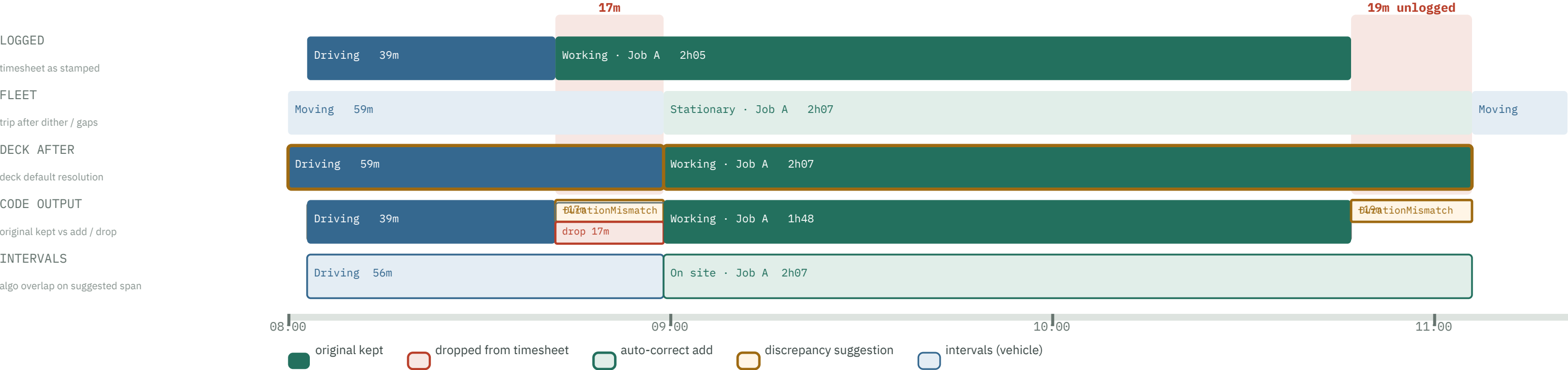
Same fleet as B1, but the technician left a gap between driving and working. Working still starts after the first onsite stop.



B3

Drive ended early and working ended before the van left the site

Clean commute then a long onsite stay. The technician closed the job about 20 minutes before the van actually left.



Cutover moves onto driving, but extra onsite time after the timesheet is only a suggestion — parked is not proof of work.

CODE MATCH

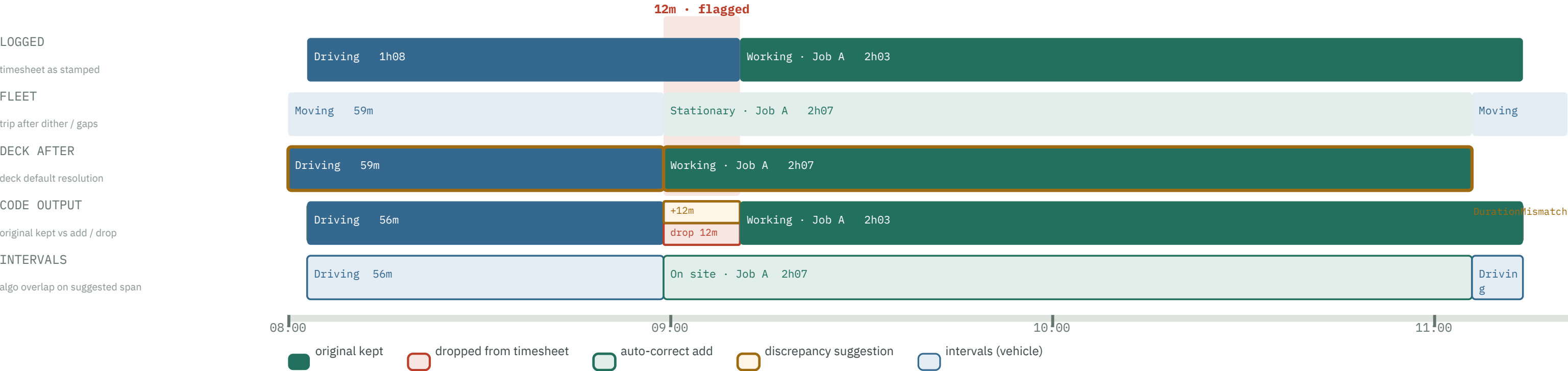
17 minutes of leftover driving are handed to the driving timesheet, but working also reaches 19 minutes into unlogged onsite time after 10:47. That reach is not a hand-over onto driving, so both sides are flagged and nothing is applied.

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Extend driving + extra onsite	17m + 19m	DISCREPANCY	DISCREPANCY

B4

Driving was logged after the van had already arrived on site

The driving timesheet runs 12 minutes into the onsite stop. Working covers the rest of the visit.



Parked-at-job offered to working is not auto-applied. Both sides of that cutover are flagged.

CODEMATCH

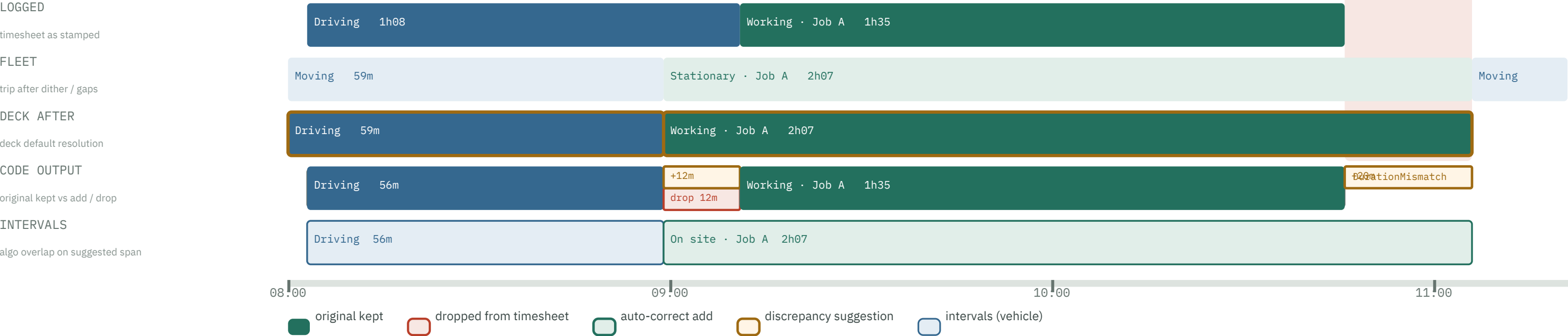
The 12-minute tail of driving that sits on the onsite stop is above the 10m floor. Parked-at-job offered to working is not auto-applied, so both sides of the cutover are flagged DurationMismatch. Times stay as logged; intervals carry the suggestion (Driving to 08:59, Working from 08:59).

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Parked ≠ working	12m	DISCREPANCY	DISCREPANCY

B5

Driving ran into the site, and working also ended early

Same late cutover as B4, and the working timesheet also ends before the van leaves.



Two reasons to flag: the cutover sits inside the onsite stop, and extra onsite time after working is only possible, not proven.

CODEMATCH

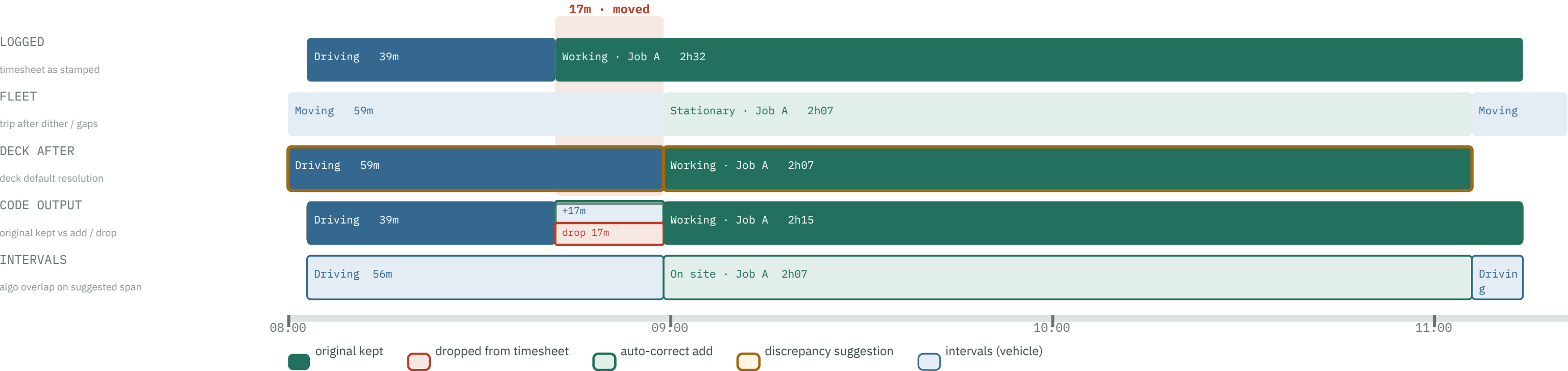
Same 12-minute late cutover as B4 (now above the floor, flagged, not applied) plus working ending 20 minutes before the van leaves. DurationMismatch; times stay as logged. Intervals show the suggested onsite window.

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Extra onsite after working	20m	DISCREPANCY	DISCREPANCY

B6

Drive ended early; working already covers the onsite window

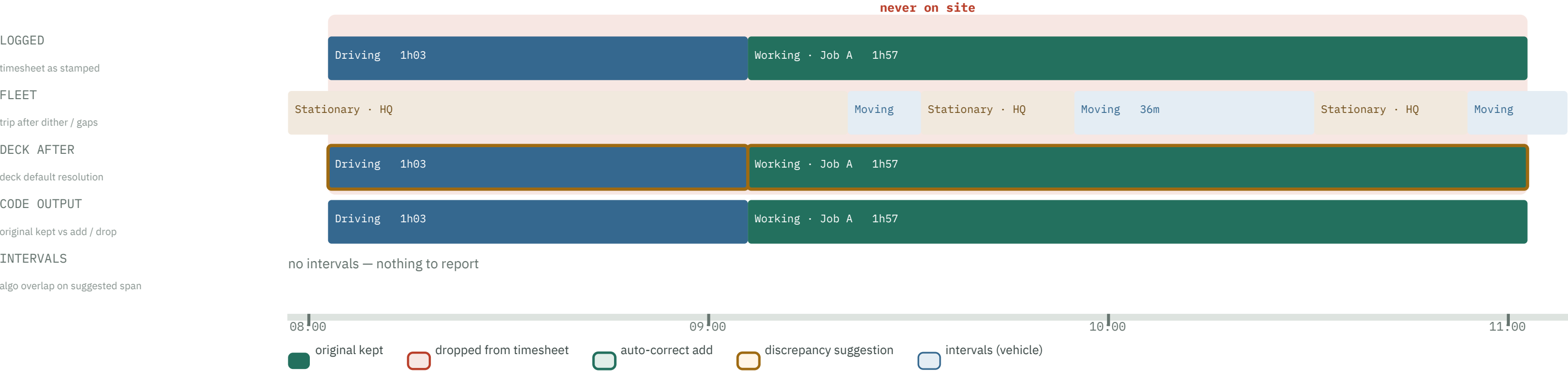
The only real mismatch is the first-drive cutover. Working already runs to the end of the visit.



B7

The van never reached Job A

Driving then working at Job A, but every fleet stop is offsite.



No onsite stop at all. Working is a missed site visit. Default resolution is gap / idle for the whole claim.

CODE

SAME TIMES, DIFFERENT LABEL

No trip matches any job site, so the whole day is skipped (likely the wrong vehicle). Empty output, times unchanged. The deck wanted a discrepancy / gap-idle for the claim. Same times, no flag.

DISCREPANCY

deck

NO ACTION

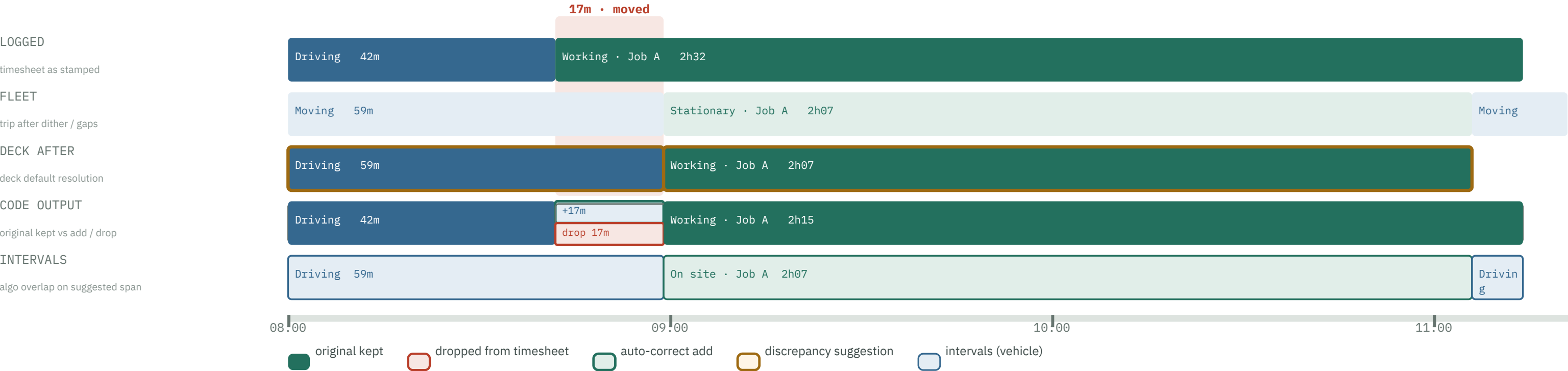
code

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Never on site	Skip day / wrong vehicle	—	DISCREPANCY	NO ACTION

B8

First drive starts with the fleet; cutover is early; working covers the visit

Logged driving begins at 08:00 with the van. The cutover is 17 minutes early; working already covers the onsite end.



Same shape as B6. Auto-edit the cutover onto driving.

CODE DIFFERS

Same auto-edit as B6: the cutover moves onto driving. Working is not trimmed for the 8-minute trailing drive-away (still under the 10m floor). Deck default ended working at 11:06; code leaves 11:14.

DETECTION

Impossible

REPLACEMENT

Extend driving

SIZE

17m

DECK OUTCOME

AUTO-EDIT

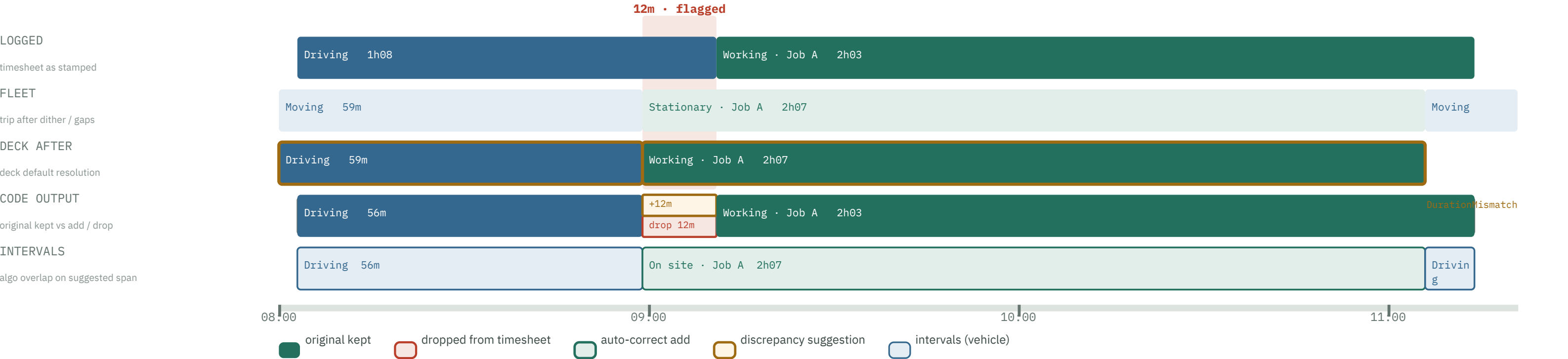
CODE OUTCOME

AUTO-EDIT

B9

Driving ran 12 minutes into the onsite stop

Same late cutover as B4, working covers the visit.



Parked time offered to working is flagged, not applied.

CODE MATCH

Identical to B4: 12 minutes of parked-at-job claimed as driving, now above the floor, flagged not applied.

DISCREPANCY

deck

DISCREPANCY

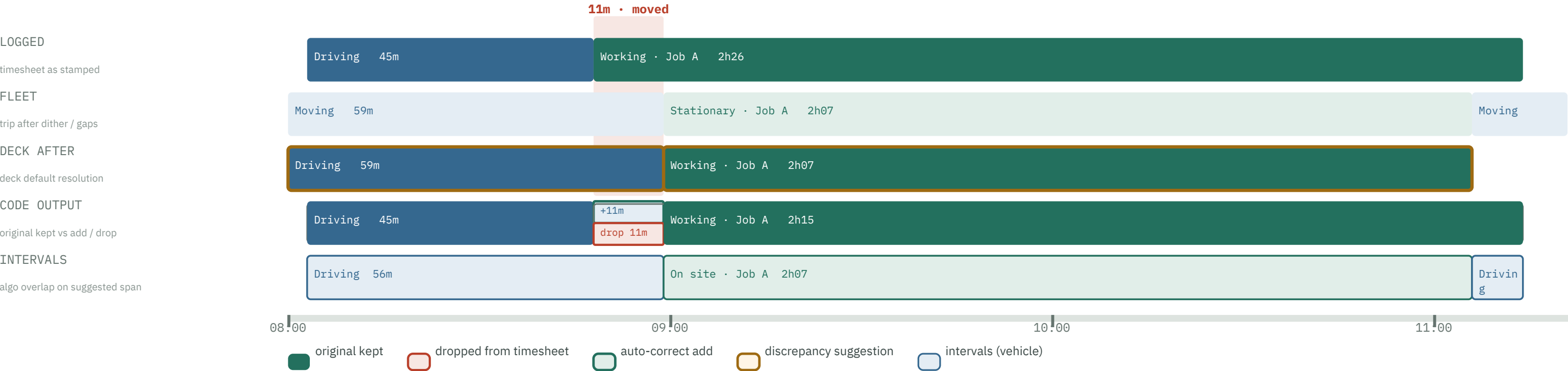
code

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Parked ≠ working	12m	DISCREPANCY	DISCREPANCY

B10

An 11-minute early cutover, working covers the visit

Drive ends at 08:48; the van arrives at 08:59. Working already runs past the onsite end.



11 minutes onto driving, under the 60m ceiling. Auto-edit.

CODE DIFFERS

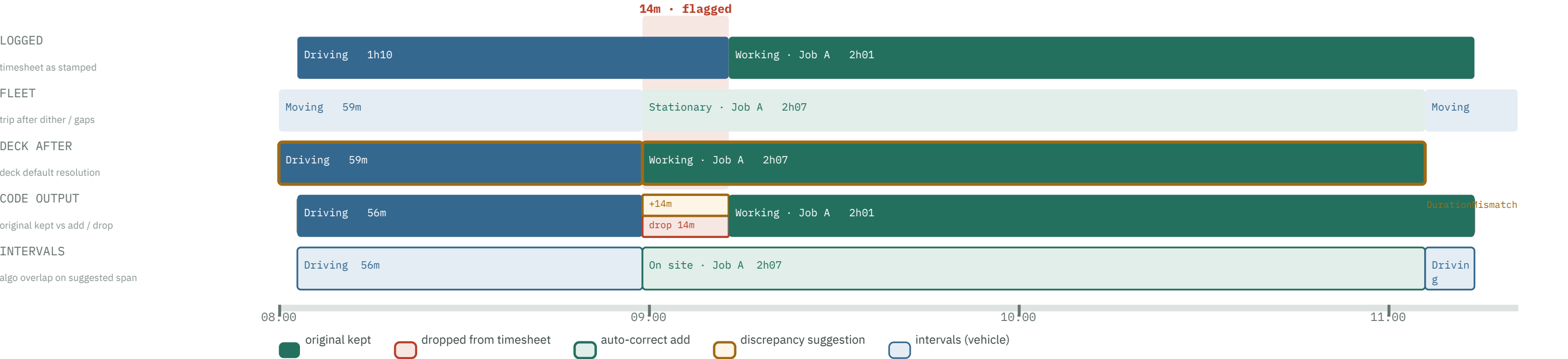
Working hands 08:48–08:59 to driving — 11 minutes, above the floor and under the ceiling, so the cutover auto-edits. The 8-minute trailing drive-away is still discarded, so working ends at 11:14 rather than the deck's 11:06.

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Extend driving	11m	AUTO-EDIT	AUTO-EDIT

B11

Driving ran 14 minutes into the onsite stop

Cutover at 09:13, van arrived at 08:59. Working covers the visit.



Same rule as B4/B9: do not auto-give parked-at-job minutes to working.

CODE MATCH

14 minutes of onsite time claimed as driving, above the floor. Parked-at-job is offered to working and flagged, not applied.

DISCREPANCY

deck

DISCREPANCY

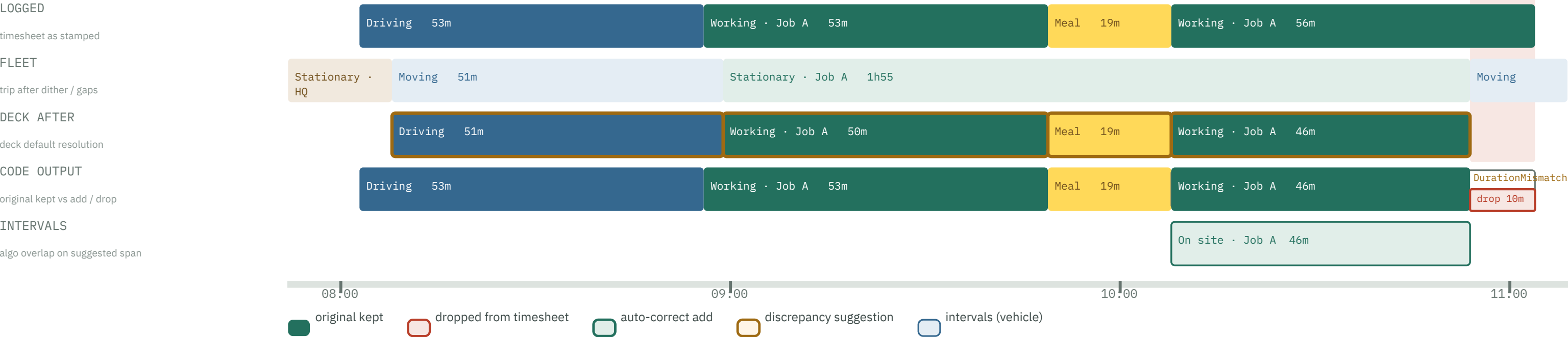
code

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Parked ≠ working	14m	DISCREPANCY	DISCREPANCY

B12

Meal break between two working timesheets at the same job

Drive, work at A, a meal, work at A again. Fleet is onsite straight through the meal.



Other/meal is left alone. Working #2 is trimmed to the last onsite stop. The first-drive cutover is offered to driving.

CODEMATCH

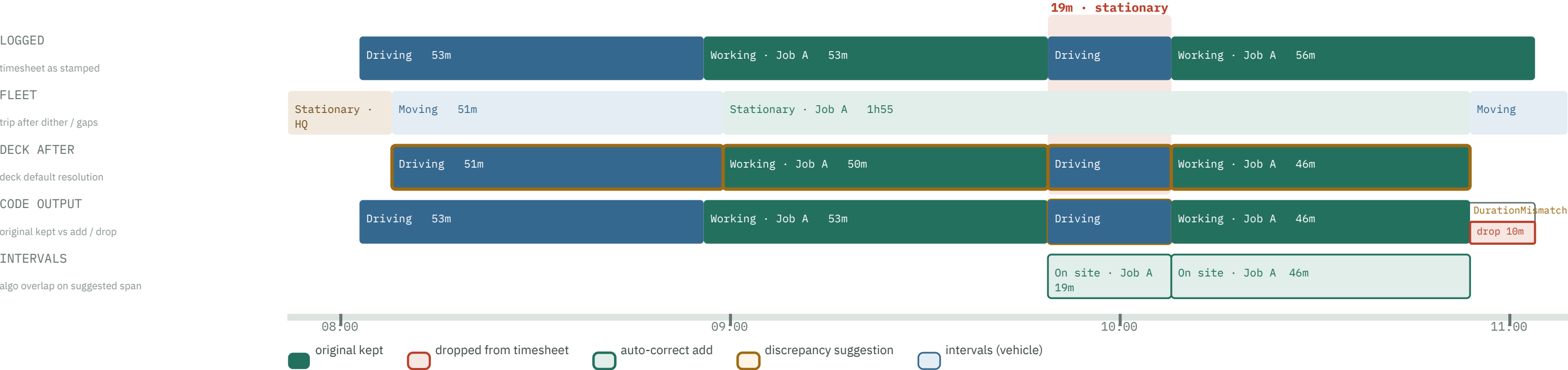
The meal is Other and is left alone. The 3-minute leftover commute is still under the floor. The 10-minute trailing drive-away on the second visit is at the floor, so that working timesheet is flagged DurationMismatch. Times stay as logged; intervals suggest ending the second visit at 10:54.

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Impossible	Trim second visit	10m	DISCREPANCY	DISCREPANCY

B13

A drive between two working timesheets at the same job

Same as B12 but the gap is logged as driving, not a meal. Fleet never left the site.



B14

Working was logged while the van sat offsite

After the commute the van stops somewhere that is not Job A for the rest of the claim.



Working never matches an onsite stop. Default resolution extends driving, then gap / idle.

CODE

SAME TIMES, DIFFERENT LABEL

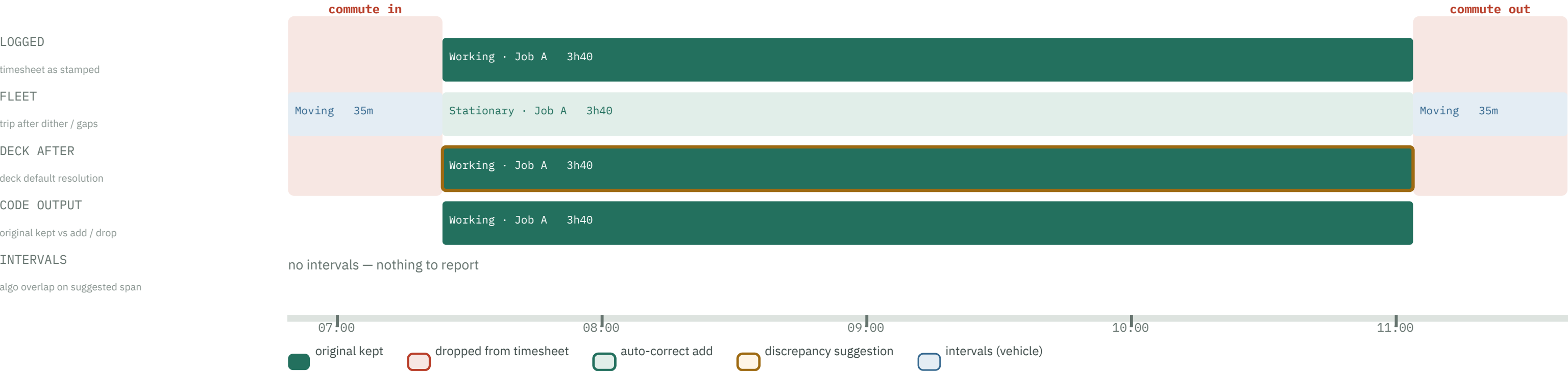
The van never matches Job A, so the day is skipped (no job-site trip). Empty output. The deck wanted driving extended and working replaced by gap / idle.

DETECTION	REPLACEMENT	SIZE	DECK OUTCOME	CODE OUTCOME
Never on site	Skip day / wrong vehicle	—	DISCREPANCY	NO ACTION

B15

Commute correctly not claimed

Only working at Job A is logged. Fleet shows a drive in, the visit, and a drive out. Add the drivings; do not add a gap.



The engine does not invent timesheets, so the unclaimed commutes are not added. Working already matches the onsite window.

CODE

SAME TIMES, DIFFERENT LABEL

Working already matches the onsite window exactly. The engine does not invent timesheets, so the unclaimed commutes are not added and nothing is flagged. The deck wanted those drivings inserted as discrepancies, with no gap.

DISCREPANCY

deck

NO ACTION

code

B16

A ten-minute visit, then driving away still logged as work

Scheduled hour at Job B. Fleet shows 10 minutes on site, then the van drives away and does not come back.

